Li and Qi as Theory and Model in Contemporary Mathematics

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Abstract

In model theory, a theory is a set of sentences, and a model is a structure in which those sentences are true. In this paper, we interpret the neo-Confucian relation between li (principle) and qi (material force) as the relation between a theory and its models, and ask what standard theorems of logic then say about the li-qi relation. The completeness theorem gives a formal counterpart to Zhu Xi's thesis that li and qi are "neither separate nor mixed." Lindström's theorem implies that a principle that fixes an infinite realization uniquely cannot be stated in a logic that supports gradual finitary inquiry. The formal framework allows us to separate the question whether li determines qi from the question of priority, with an emphasis on the former question, and so clarifies the historical positions of Zhu Xi, Wang Fuzhi, T'oegye and Yulgok.

Keywords: li-qi; neo-Confucianism; logic; model theory; categoricity

1 Introduction

The relationship between li (principle, pattern) and qi (material force, vital stuff) was one of the main metaphysical questions of neo-Confucian philosophy for roughly seven centuries, first in China and later in Korea, where the debate took distinctive forms. The debate

concerned the priority of li over qi and the separability of li from qi. Priority asks whether li comes before qi, and in what sense. Separability asks whether li can be anything apart from the order found in qi. Zhu Xi held that li and qi are never found apart and yet are not the same thing. He also held that li is prior to qi in a logical sense, with no temporal order between them (Zhu Xi [1270] 1986; Chan 1963). By contrast, Wang Fuzhi held that li is the pattern of qi and has no standing apart from qi (Wang Fuzhi 2011). In Korea the Four-Seven controversy between Yi Hwang (T'oegye) and Ki Taesŭng (Kobong) sharpened the debate, with Yi I (Yulgok) later entering the debate with a separate view (Kalton et al. 1994; Chung 1995; Ro 1989).

This paper proposes a framework for interpreting this debate, drawn from model theory, the branch of mathematical logic that studies the relation between formal languages and the structures that interpret them (Chang and Keisler 1990; Hodges 1993; Marker 2002). Model theory distinguishes a theory from a model. A theory is a set of sentences in a formal language, and a model of the theory is a mathematical structure in which all of those sentences are true. We argue that the relation between a theory and its models corresponds closely to what the neo-Confucians said about li and qi, and that the major theorems of model theory can be read as statements about how far a principle can determine its realizations.

For modern treatments of the metaphysics of li and qi, see Angle and Tiwald (2017) and Liu (2017). Adler (2023) discusses Zhu Xi's synthesis, and Kim and Kim (2023) discuss Confucianism in Chosŏn Korea. Button and Walsh (2018) survey the philosophy of model theory.

Given the correspondence in Table 1, each theorem we discuss has a definite translation into the vocabulary of li and qi. Whether a translated statement clarifies a particular neo-Confucian thesis is an interpretive question, which we argue case by case. We do not claim that neo-Confucian thinkers anticipated model theory, nor that model theory settles their debates. We only claim that model theory supplies a vocabulary in which some of their disagreements become sharper, and in which some apparent disagreements turn out to be

about different questions.

Classical model theory studies static structures, whereas qi is dynamic and generative. We set aside the temporal dynamics of qi and consider only the logical relation between a principle and its instantiations. Section 11 returns to this limitation.

Section 2 gives the background in model theory and in neo-Confucian thought. Section 3 sets out the correspondence and distinguishes an axiomatic reading of li, in which li is a theory, from a structural reading, in which li is the complete theory of a given structure. Section 3 also replaces the traditional pair of questions, priority and separability, with priority and determination. Determination asks whether li fixes qi up to isomorphism, that is, up to a renaming of objects that preserves all structure. Most of the mathematics in this paper concerns determination. Section 4 reads the completeness theorem as a counterpart of Zhu Xi's "neither separate nor mixed." Section 5 argues from Lindström's theorem and the categoricity of second-order arithmetic that a principle cannot both determine an infinite realization and be grasped piece by piece. Section 6 uses Morley's theorem and standard examples to show that determination comes in degrees, and Section 7 turns to non-standard arithmetic and incompleteness. Section 8 restates the argument from Tennenbaum's theorem and sets out the limits of that argument. Section 9 proposes that the model-theoretic notion of a type gives an analogue of the neo-Confucian distinction between original and physical nature and of Yulgok's thesis that li pervades while qi delimits. Section 10 places the historical positions with respect to priority and determination, and Section 11 considers objections.

2 Background

2.1 Model theory in brief

We fix only the notation we need. Chang and Keisler (1990), Hodges (1993) and Marker (2002) give full treatments.

Definition 1 (Language, structure, satisfaction)**.** A language $\mathcal{L}$ is a set of constant, function

and relation symbols. An $\mathcal{L}$-structure M consists of a non-empty set D, called the domain, together with an interpretation of each symbol of $\mathcal{L}$ as an element, function or relation on D. For an $\mathcal{L}$-sentence φ we write $M \models \varphi$ if φ is true in M, and for a set of sentences T we write $M \models T$ if $M \models \varphi$ for every $\varphi \in T$.

Definition 2 (Theory, model, complete theory)**.** A theory is a set T of $\mathcal{L}$-sentences. A model of T is a structure M with $M \models T$, and $\mathrm{Mod}(T)$ denotes the class of all models of T. The complete theory of a structure M is $\mathrm{Th}(M) = \{\varphi : M \models \varphi\}$. Two structures M and M' are elementarily equivalent, written $M \equiv M'$, if $\mathrm{Th}(M) = \mathrm{Th}(M')$, that is, if the same sentences are true in both. We write $T \vdash \varphi$ if φ can be derived from T by a formal proof. A theory T is consistent if no contradiction can be derived from T, and complete if $T \vdash \varphi$ or $T \vdash \neg\varphi$ for every sentence φ.

Definition 3 (Isomorphism and categoricity)**.** Two structures are isomorphic if a one-to-one correspondence between their domains preserves all constants, functions and relations. Isomorphic structures differ only in the names of their elements. A theory is categorical if all its models are isomorphic. The cardinality of a structure is the number of elements in its domain, which may be finite, countably infinite or uncountable. For an infinite cardinal κ, a theory is κ-categorical if all its models of cardinality κ are isomorphic.

The maps $T \mapsto \mathrm{Mod}(T)$ and $M \mapsto \mathrm{Th}(M)$ run in opposite directions, from sentences to structures and from structures to sentences. Section 10 uses these maps to describe explanation that runs from li to qi and explanation that runs from qi to li.

2.2 Neo-Confucian background

Several neo-Confucian theses recur in this paper. Translations follow Chan (1963) and Kalton et al. (1994) unless otherwise indicated.

Zhu Xi held that li and qi are "neither separate nor mixed" (bu li bu za). There is no li without qi and no qi without li, and yet li and qi are not the same thing and must not be

confused. Zhu Xi says that without qi, li would have "nothing to adhere to." Asked which comes first, he answers that li and qi have no temporal order, but that li is prior if one insists on an order. Zhu Xi also holds that the li of a thing can be present before any such thing exists: "before there were ruler and minister, there was already the principle of ruler and minister" (Zhu Xi [1270] 1986, our translation).

The thesis that principle is one while its manifestations are many (li yi fen shu) goes back to Cheng Yi, and Zhu Xi developed the thesis further. A single li is wholly present in each thing, and things differ because their endowments of qi differ. We call the resulting question the "one li, many qi" problem.

In Korea the Four-Seven debate concerned the Four Beginnings and the Seven Feelings. The Four Beginnings are the moral feelings named by Mencius: commiseration, shame, deference, and the sense of right and wrong. The Seven Feelings are the emotions listed in the Book of Rites, such as joy, anger and sorrow. T'oegye held that the Four Beginnings issue from li, with qi following, and that the Seven Feelings issue from qi, with li riding on qi. This view is known as the doctrine of mutual issuance.

Yulgok entered the Four-Seven debate later, in an exchange of letters with Sŏng Hon (Ugye). In that exchange Yulgok rejected T'oegye's mutual issuance and held that only qi "issues," or acts, and that li "rides" on qi. Yulgok also held that li pervades while qi delimits (li tong qi ju). In Yulgok's view li is universal and unobstructed, qi is particular and limiting, and all differentiation among things comes from qi (Kalton et al. 1994; Chung 1995; Ro 1989).

3 The correspondence and its scope

3.1 The choice of model theory

Model theory studies the relation between syntax and semantics (Hodges 1993). Syntax consists of symbols and rules for forming sentences, and the symbols carry no content until they are interpreted. Syntax corresponds to li considered apart from things, which Zhu Xi

described as without form (Zhu Xi [1270] 1986). Semantics assigns the symbols to a domain of objects and corresponds to the instantiation of li in concrete qi. A theory determines what any structure satisfying the theory must be like, but the theory is not such a structure. Similarly, li determines the nature of things without being a thing.

We call the correspondence in Table 1 an analogy. We avoid the word "isomorphism" for the correspondence, because the word has a technical meaning in model theory, and the neo-Confucian concepts do not form a mathematical structure on which an isomorphism could be defined.

Table 1: Correspondence between neo-Confucian metaphysics and mathematical logic.

Concept	**Neo-Confucian**	**Logic**	**Attributes**
Principle	li	Theory T	Abstract, axiomatic, rule-giving, logically prior
Material force	qi	Model M	Concrete, structured, populated by objects
Relation	li-qi	Satisfaction $M \models T$	The structure (qi) realizes the axioms (li)
Li of a thing	li in qi	$\mathrm{Th}(M)$	The full pattern read off from one realization
Endowment	qizhi	Types realized in M	Which possibilities a given qi manifests (Sec. 9)

3.2 Two readings of li

The correspondence allows an axiomatic and a structural reading of li. The axiomatic reading takes li to be a theory T, usually given by a small set of axioms that an algorithm can list, and lets qi range over $\mathrm{Mod}(T)$. This reading presents li as something that can be studied and taught, a principle that someone could write down. The structural reading takes li to be $\mathrm{Th}(M)$, the totality of what is true in a particular structure M. This reading presents li as wholly present in the thing and complete whether or not anyone grasps the principle.

The structural reading answers an obvious objection to our framework, namely that li is not linguistic. $\mathrm{Th}(M)$ is determined by M alone. $\mathrm{Th}(M)$ is the pattern of M as far as

a given language can register that pattern, and $\mathrm{Th}(M)$ does not depend on any particular choice of axioms.

Note that the axiomatic and structural readings differ in the completeness of a theory. $\mathrm{Th}(M)$ is always complete, because every sentence or the negation of the sentence is true in M, while an axiomatic theory often leaves some sentences undecided. Section 7 shows that the complete theory of a simple structure such as the natural numbers need not have any set of axioms that an algorithm can list.

3.3 Determination and priority

The traditional debate concerns priority and separability. The theorems in this paper concern determination, so we pair priority with determination instead of separability. Determination is the question whether a given li fixes its qi up to isomorphism. Mathematically, the question is whether T is categorical, or κ-categorical for the relevant κ. Priority is the question which of li and qi is explanatorily prior. Mathematically, the question is whether we start from T and study $\mathrm{Mod}(T)$, or start from M and study $\mathrm{Th}(M)$.

Determination and priority are independent questions. A dualist who holds that li is prior can accept that one li has many qi, since li yi fen shu affirms that claim. A monist who holds that li is nothing but the order of qi can accept the same multiplicity by treating the other models as mathematical possibilities without physical reality. A unique qi is also compatible with both views. For this reason a result about determination is not by itself an argument for or against dualism.

4 Completeness: no consistent li without qi

We begin with the completeness theorem, which links the consistency of a set of sentences to the existence of a structure in which the sentences are true. This link is the formal analogue of the claim that every coherent li has a qi.

Theorem 1 (Completeness theorem; Gödel 1930)**.** A first-order theory T is consistent if and only if T has a model. Equivalently, for every sentence φ, $T \vdash \varphi$ if and only if φ is true in every model of T.

Interpretation 1. The completeness theorem says that every coherent li is realized in some qi. A "principle" with no possible realization would be inconsistent, and an inconsistent set of sentences is not a principle at all. This result is a formal counterpart of the clause "not separate" in Zhu Xi's thesis. The clause "not mixed" corresponds to the fact that a theory and a model are different kinds of object. A theory consists of sentences, a model is a structure, and the two are connected only by the relation $\models$.

The completeness theorem guarantees a mathematical model, but the model need not be physically actual. This fits Zhu Xi's claim that the li of ruler and minister precedes any actual ruler or minister, since a theory can have models without any of the models being realized in the world we inhabit. The completeness theorem therefore supports the independence of li from actual qi and denies the independence of li from possible qi.

5 Compactness, Löwenheim-Skolem, and the price of uniqueness

5.1 Compactness

The compactness theorem explains why a first-order principle can be investigated in finite pieces.

Theorem 2 (Compactness)**.** A first-order theory has a model if and only if every finite subset of the theory has a model.

Suppose a candidate li is an infinite set of axioms, and we can inspect only finitely many of the axioms at a time. Compactness says that if each finite fragment is realizable in some qi, possibly in a different qi for each fragment, then the whole li is realizable in a single qi.

Interpretation 2. Compactness guarantees the coherence of li and says nothing about the reliability of induction. Zhu Xi's program of investigating things (gewu), in which principle is grasped piece by piece until the whole is understood, presupposes that the pieces fit together. Compactness assures that a first-order li cannot hide a contradiction only at infinity. If the whole li has no qi, some finite part of the li already has none.

5.2 The Löwenheim-Skolem theorems

The Löwenheim-Skolem theorems show that a first-order theory cannot control the size of its models. A substructure M_0 of M is elementary if every sentence with parameters from M_0 has the same truth value in M_0 as in M, and M is then called an elementary extension of M_0.

Theorem 3 (Löwenheim-Skolem)**.** Let T be a theory in a countable language, and let M be an infinite model of T. For every infinite cardinal $\kappa < |M|$, M has an elementary substructure of cardinality κ (downward theorem). For every cardinal $\kappa > |M|$, M has an elementary extension of cardinality κ (upward theorem). Consequently, T has a model of every infinite cardinality.

A set is countable if the elements of the set can be listed as $0, 1, 2$, and so on. The downward theorem (Skolem 1920) says that any infinite qi realizing a li contains a countable qi in which the same sentences are true. The actual world may be vast, but every first-order truth about the world already holds in a countable part of the world.

The upward theorem has a consequence for determination.

Corollary 1. No first-order theory with an infinite model is categorical.

Proof. Models of different cardinalities cannot be isomorphic, and by the upward Löwenheim-Skolem theorem such a theory has models of arbitrarily large cardinality. □

Corollary 1 states the "one li, many qi" result in general form. Any first-order li with an infinite qi admits other qi that are not isomorphic to that qi.

5.3 Lindström's theorem and second-order logic

One might hope to avoid Corollary 1 by moving to a more expressive logic. Lindström's theorem shows the cost of such a move. The theorem concerns abstract logics, which generalize first-order logic by allowing other ways of forming sentences, such as quantifiers over sets. A logic is regular if the logic satisfies basic closure conditions, such as closure under the Boolean connectives and under renaming of symbols. A logic has the compactness property if every set of sentences whose finite subsets all have models has a model. A logic has the downward Löwenheim-Skolem property if every sentence with an infinite model has a countable model.

Theorem 4 (Lindström 1969)**.** Let Λ be a regular abstract logic that extends first-order logic. If Λ has both the compactness property and the downward Löwenheim-Skolem property, then Λ is no more expressive than first-order logic.

Second-order logic illustrates the trade-off. In second-order logic, quantifiers may range over all subsets of the domain. Dedekind showed that second-order Peano arithmetic PA_2, whose induction axiom quantifies over all subsets, is categorical under the standard semantics, in which the second-order quantifiers range over all subsets. Every model of PA_2 is isomorphic to the natural numbers $\mathbb{N}$ (Dedekind 1888; Shapiro 1991). One li thus determines one qi. By Lindström's theorem, second-order logic must lose compactness or the Löwenheim-Skolem property, and in fact second-order logic loses both.

Second-order logic also has no complete proof system in which an algorithm can check proofs. If such a system existed, an algorithm could list all consequences of the single sentence PA_2. Because PA_2 is categorical, the first-order sentences among these consequences coincide with true arithmetic $\mathrm{Th}(\mathbb{N})$, the set of all first-order sentences true in $\mathbb{N}$. An algorithm could then list $\mathrm{Th}(\mathbb{N})$, and because $\mathrm{Th}(\mathbb{N})$ is complete, an algorithm could also decide membership in $\mathrm{Th}(\mathbb{N})$. Gödel's incompleteness theorem rules this out. If second-order quantifiers are instead taken to range over some subsets of the domain (Henkin semantics; Henkin 1950), completeness and compactness return, and non-standard models return with them.

Interpretation 3 (The price of uniqueness)**.** In any regular logic with both compactness and the downward Löwenheim-Skolem property, no theory with an infinite model determines that model up to isomorphism. Lindström's theorem shows that such a logic is no more expressive than first-order logic, and Corollary 1 then applies. A li that determines an infinite qi must therefore be stated in a logic that lacks one of these properties, and second-order logic also lacks a complete method of proof. We call this trade-off between determinacy and accessibility the price of uniqueness. Finite qi are an exception, because the first-order theory of a finite structure has only that structure as a model, up to isomorphism.

The price of uniqueness offers an interpretation of a tension in Zhu Xi's epistemology. The investigation of things proceeds incrementally, one principle at a time. In his supplement to the commentary on the Great Learning, Zhu Xi nonetheless describes the completion of this investigation as a sudden, comprehensive penetration (huoran guantong), in which the whole is grasped at once (Chan 1963). Lindström's theorem suggests why incremental investigation and sudden penetration might differ. Principles accumulated one at a time behave like the axioms of a first-order theory, and by Corollary 1 a first-order theory does not determine an infinite model. A principle that determines an infinite realization, such as PA_2, belongs to a logic that is not compact. In such a logic a set of principles can lack a model even though every finite part of the set has one. Second-order logic also has no complete proof system through which all consequences of PA_2 could be derived step by step. If the sudden penetration is a grasp of a principle that fixes a unique realization, the penetration cannot be the sum of the incremental steps.

6 A spectrum of determination

Corollary 1 rules out categoricity for first-order theories with infinite models. However, uniqueness among models of a fixed size remains possible. This weaker property is κ-categoricity, and κ-categoricity comes in degrees. Morley's theorem shows that a complete

theory in a countable language with a unique model at one uncountable size has a unique model at every uncountable size.

Theorem 5 (Morley 1965)**.** Let T be a complete theory in a countable language. If T is κ-categorical for some uncountable κ, then T is κ-categorical for every uncountable κ.

Standard examples show the range of possibilities (Marker 2002). Consider the infinite vector spaces over a fixed finite field, such as the field with two elements. The theory of these vector spaces is categorical in every infinite cardinality, so once the size of qi is fixed, li determines qi completely.

A dense linear order without endpoints is an ordering in which any two elements have a third element between them and in which there is no least or greatest element. The rational numbers form such an order. Cantor's back-and-forth argument shows that any two countable dense linear orders without endpoints are isomorphic, so the theory of these orders is $\aleph_0$-categorical. The theory also has 2^κ non-isomorphic models in every uncountable cardinality κ, which is the largest number possible (Shelah 1990). The li of dense order thus fixes qi at the countable size and fixes almost nothing at larger sizes.

Algebraically closed fields of characteristic zero show the reverse pattern. The complex numbers form such a field. The theory of these fields is categorical in every uncountable cardinality but has countably many countable models. The countable models are classified by transcendence degree, the number of elements that are algebraically independent over the rational numbers, and the transcendence degree can be $0, 1, 2$, and so on, or $\aleph_0$.

True arithmetic $\mathrm{Th}(\mathbb{N})$ shows the most variety. $\mathrm{Th}(\mathbb{N})$ has $2^{\aleph_0}$ non-isomorphic countable models and is categorical in no infinite cardinality (Shelah 1990).

Interpretation 4. Whether one li admits one qi or many depends on the li and on the size of qi. Morley's theorem says that for a complete li in a countable language, behavior at uncountable sizes is uniform. If such a li determines qi at one uncountable size, the li determines qi at every uncountable size. Neo-Confucian discussions of li yi fen shu were not

concerned with cardinality, but they were concerned with the manner in which one principle is distributed among many things. The examples show that this distribution can take quite different forms.

7 Arithmetic: non-standard qi and the limits of effective li

7.1 Non-standard models of Peano arithmetic

Arithmetic is the most familiar case of one li with many qi. First-order Peano arithmetic PA is formulated in the language $\mathcal{L}_{\mathsf{PA}} = \{0, 1, +, \times, <\}$. The axioms of PA describe the successor structure, addition, multiplication and order, together with the induction schema. The induction schema says that if a property defined by a formula holds of 0 and passes from each number to the next, then the property holds of every number. Formally, for each formula $\varphi(x, \bar{y})$,

$$\forall \bar{y} \left[\Big(\varphi(0, \bar{y}) \wedge \forall x \left(\varphi(x, \bar{y}) \rightarrow \varphi(x+1, \bar{y}) \right) \Big) \rightarrow \forall x \, \varphi(x, \bar{y}) \right]. \tag{1}$$

The second-order induction axiom quantifies over all subsets $P \subseteq \mathbb{N}$; by contrast, the first-order schema covers only the subsets that a formula can define. There are only countably many formulas but $2^{\aleph_0}$ subsets of $\mathbb{N}$, so the first-order schema constrains a vanishingly small fraction of the properties that the second-order axiom constrains.

Non-standard models of PA exploit this gap. A non-standard model of PA is a model that is not isomorphic to $\mathbb{N}$. Such a model contains copies of the standard numbers $0, 1, 2, \ldots$ and also elements larger than all of the standard numbers. The following proposition shows that such models exist.

Proposition 1. PA has a countable model not isomorphic to $\mathbb{N}$.

Proof. Add a new constant c to the language and let $\underline{n}$ abbreviate the term $1 + \cdots + 1$ (n times). Let

$$T' = \mathsf{PA} \cup \{\, \underline{n} < c : n \in \mathbb{N} \,\}.$$

Any finite subset of T' mentions only finitely many $\underline{n}$, and $\mathbb{N}$ satisfies the finite subset when c is interpreted as a sufficiently large number. By compactness T' has a model, and by the downward Löwenheim-Skolem theorem T' has a countable model M. The element c^M lies above every standard numeral, so M is not isomorphic to $\mathbb{N}$, and $M \models \mathsf{PA}$. □

7.2 Incompleteness: the gap between effective li and complete li

The incompleteness theorem shows that no axiom system that an algorithm can list proves every arithmetical truth. A theory is recursively axiomatizable if an algorithm can list the axioms of the theory. Robinson arithmetic Q is a weak fragment of PA with finitely many axioms and no induction.

Theorem 6 (Gödel-Rosser; Gödel 1931; Rosser 1936)**.** Let T be a consistent, recursively axiomatizable theory that interprets Robinson arithmetic Q. Then T is incomplete: there is a sentence φ such that $T \nvdash \varphi$ and $T \nvdash \neg\varphi$.

One of φ and $\neg\varphi$ is true in $\mathbb{N}$, so any such T that is true in $\mathbb{N}$ fails to prove some truth about $\mathbb{N}$. One might conclude that qi has a semantic richness that li cannot exhaust. The axiomatic and structural readings of li from Section 3 qualify this conclusion. The structural reading takes the li of $\mathbb{N}$ to be $\mathrm{Th}(\mathbb{N})$, which is complete by definition and omits nothing. Incompleteness shows that $\mathrm{Th}(\mathbb{N})$ is not recursively axiomatizable, because no set of principles that an algorithm can list yields $\mathrm{Th}(\mathbb{N})$.

Interpretation 5. Incompleteness reveals a gap between the complete li of $\mathbb{N}$ and any li that an algorithm can list. The relation between li and qi is not at issue. This reading of incompleteness is closer to Zhu Xi's view than the conclusion that qi is richer than li. For Zhu Xi, li is complete and fully present in each thing, but human understanding of li is partial and

must be cultivated. Incompleteness shows that no curriculum of principles that an algorithm can list will ever be complete.

8 Tennenbaum's theorem: mechanism and vitalism

8.1 The theorem and the basic argument

Tennenbaum's theorem connects the uniqueness of a model of arithmetic to computability. A countable model is computable if the elements of the model can be numbered $0, 1, 2$, and so on, so that an algorithm computes the operations of the model on these numbers. Such a numbering is called a presentation of the model with domain $\mathbb{N}$.

Theorem 7 (Tennenbaum 1959)**.** If M is a countable non-standard model of PA, then neither $+^M$ nor $\times^M$ is computable, under any presentation of M with domain $\mathbb{N}$. Consequently, up to isomorphism, $\mathbb{N}$ is the only computable model of PA.

Tennenbaum's theorem suggests a dichotomy between mechanical and vital qi. If qi is mechanical, in the sense that an algorithm can generate the countable realization of arithmetic in qi, then the li of arithmetic picks out a single qi, the standard natural numbers. If qi is instead vital, with features that no algorithm can compute, non-standard models are no longer excluded and one li again admits many qi.

8.2 Limits of the argument

The argument has limits. Tennenbaum's theorem depends on the strength of PA, and weaker theories of arithmetic can have computable non-standard models. Shepherdson (1964) constructed a computable non-standard model of arithmetic with induction restricted to open formulas, which are formulas without quantifiers. Whether mechanical qi forces uniqueness therefore depends on how rich the governing li is. A richer principle leaves fewer computable

realizations. Open induction allows computable non-standard models, PA allows only $\mathbb{N}$, and set theory, discussed below, allows none.

Tennenbaum's theorem also depends on the choice of primitive symbols. Two theories are definitionally equivalent when each theory can define the primitive notions of the other. Definitionally equivalent theories prove the same things once one theory is translated into the other, and they differ only in which notions they take as basic. Pakhomov (2022) constructs a theory that is definitionally equivalent to PA and that has a computable non-standard model, unlike PA in the usual symbols $\{0, 1, +, \times, <\}$. Tennenbaum's theorem is not preserved under definitional equivalence. Whether a non-standard model can be computable depends on how the theory is written, and not only on what the theory says. Pakhomov's method also yields a theory definitionally equivalent to Zermelo-Fraenkel set theory (ZF) that has a computable model. Pakhomov also proves a limit to the method: no theory definitionally equivalent to true arithmetic $\mathrm{Th}(\mathbb{N})$ has a computable non-standard model.

Pakhomov's results separate the axiomatic and structural readings of li from Section 3. If li is an axiomatic theory, whether a computable qi is unique can depend on the vocabulary in which li is written. If li is the complete theory $\mathrm{Th}(\mathbb{N})$, the vocabulary makes no difference. The Tennenbaum argument is therefore stronger under the structural reading, which is also the reading that answers the objection that li is not linguistic (Section 11).

Tennenbaum's argument also has a limited scope. Zermelo-Fraenkel set theory with the axiom of choice (ZFC), written in the usual vocabulary with the membership symbol $\in$ alone, has no computable models, and much weaker set theories have none either. Rabin (1958) gives a related argument, and Godziszewski and Hamkins (2017) prove a stronger result. No model of ZFC, or of considerably weaker set theories, has even a computable quotient presentation, in which an algorithm computes the operations on numbers that may stand for the same element. If the li of the world resembles ZFC more than PA, then the mechanical hypothesis does not single out one model in the usual vocabulary. The hypothesis rules out every model, and so the hypothesis cannot be held together with that li.

Tennenbaum's theorem says nothing about uncountable models, for which computability in the usual sense is not defined. If qi is modeled as a continuum, as in most physical theories, the argument does not apply without modification.

A further limit concerns circularity. Computability is defined using the standard natural numbers. A function is computable if a Turing machine computes the function, and the steps of a computation are counted in $\mathbb{N}$. Using computability to single out the standard model may therefore assume that we already know which model is the standard one. Button and Walsh (2018) discuss this objection when they assess whether Tennenbaum's theorem answers Putnam's model-theoretic argument against realism (Putnam 1980). The mechanical hypothesis about qi thus already contains a li of number, and the hypothesis cannot provide an independent route from li to a unique qi.

Interpretation 6. Because of these limits, the Tennenbaum argument establishes a conditional and local result. If the arithmetical li is sufficiently rich and the arithmetical part of qi is computable relative to the standard numbers, the arithmetical part of qi is unique up to isomorphism. This result is surprising, but the result does not by itself pick whether mechanism or vitalism is true for qi, and still less whether monism or dualism is true.

9 Types: original nature, physical nature, and "li pervades, qi delimits"

Earlier sections ask whether models of the same theory are isomorphic. The notion of a type instead describes how models of the same theory differ, and we suggest that types give an analogue of the neo-Confucian distinction between original nature and physical nature.

A sentence is either true or false in a model: the truth value is fully determined. A formula with a free variable x, such as $x > \underline{2}$, is instead true for some elements of a model and false for others. The collection of formulas that are true for a given element provides a full description of that element in the language of a theory. A type is this description but

considered without referring to a particular model or an element in the model. The type lists everything that can consistently be said about one possible element, given T, and a model may or may not contain an element that fits the description.

Definition 4 (Types)**.** Let T be a complete theory, so that T decides every sentence of the language. A complete n-type of T is a set $p(x_1, \ldots, x_n)$ of formulas in the free variables $x_1, \ldots, x_n$ that is consistent with T and maximal, in the sense that for every such formula φ, either φ or $\neg\varphi$ belongs to p. A model $M \models T$ realizes p if some tuple $\bar{a}$ of elements of M satisfies every formula in p, and M omits p otherwise.

Arithmetic gives the simplest examples. In $\mathbb{N}$ the number 2 satisfies the formula $x = \underline{2}$, and the type of 2 is the set of all formulas that are true of 2 in $\mathbb{N}$. Every model of $\mathrm{Th}(\mathbb{N})$ realizes this type, because $\mathrm{Th}(\mathbb{N})$ says that some element satisfies $x = \underline{2}$ and also proves that any such element satisfies every other formula in the type. A type with this property is called isolated: a single formula in the type implies all the other formulas in the type, and every model of T realizes every isolated type. The formulas $x > \underline{n}$ for $n = 0, 1, 2$, and so on, describe a different kind of element, one larger than every standard number. By compactness these formulas are consistent with $\mathrm{Th}(\mathbb{N})$, and they extend to complete types that are not isolated. Non-standard models of $\mathrm{Th}(\mathbb{N})$, built as in Proposition 1, realize such types, and $\mathbb{N}$ omits all of them, because every element of $\mathbb{N}$ is a standard number.

These examples illustrate general facts (Marker 2002). By compactness every type of T is realized in some model of T, so a type is always a real possibility for the theory. A model realizes every isolated type but need not realize the others. The omitting types theorem shows that any single non-isolated type of a complete theory in a countable language is omitted by some countable model of the theory. Some theories have a prime model, which realizes only the isolated types, and $\mathbb{N}$ is the prime model of $\mathrm{Th}(\mathbb{N})$. At the other extreme, an ω-saturated model realizes every type of T, and every complete theory with infinite models has ω-saturated models.

Interpretation 7. Neo-Confucians distinguished the original nature (benran zhi xing) from

the physical nature (qizhi zhi xing). The original nature is the li that is identical and complete in every person. The physical nature differs from person to person according to the clarity or turbidity of the person's endowment of qi. Turbid qi obscures li without removing li, so that li is present but not manifest. All models of a complete theory share the same li, since the same sentences are true in every model, and yet each model realizes some types and omits others. The analogy can be read in more than one way, depending on what it means for li to be manifest. If manifesting li means realizing everything consistent with li, an omitted type is a real possibility of li that the model does not manifest. For example, the standard model $\mathbb{N}$ omits every type of an element larger than all standard numbers, although such types are consistent with the li of arithmetic. An ω-saturated model realizes every type and then corresponds to perfectly clear qi, in which li is manifest without obstruction. If manifesting li instead means realizing nothing beyond what li fixes, things differ significantly. Every element of $\mathbb{N}$ is named by a numeral, so every type that $\mathbb{N}$ realizes is isolated, and $\mathbb{N}$ embeds as an elementary substructure into every model of $\mathrm{Th}(\mathbb{N})$. The part of each model that li determines is then present in every model, and the non-isolated types of non-standard elements are additions that obscure li without removing li. Section 5 supports this reading, because non-standard elements arise from the limits of first-order logic, and the categorical second-order theory PA_2 admits no model that contains them. The standard model $\mathbb{N}$ then corresponds to clear qi, and an ω-saturated model to turbid qi. This reading requires a prime model, which not every complete theory has, while ω-saturated models always exist. The choice between the readings depends on whether the clear qi of the sage is thought to manifest every possibility consistent with li or only what li fixes.

The same framework fits Yulgok's thesis that li pervades and qi delimits. The theory is identical in every model of a complete theory T, while size, realized types and isomorphism type vary from model to model. Yulgok's thesis likewise holds that li is universal and unobstructed, and that all particularity comes from qi.

10 Historical positions reframed

This section places the main historical positions with respect to priority and determination.

Zhu Xi takes li to be logically prior, which corresponds to starting from T and studying $\mathrm{Mod}(T)$. The thesis li yi fen shu already accepts that one li has many qi, so Corollary 1 is consistent with Zhu Xi's view. The completeness theorem corresponds to his thesis that li and qi are neither separate nor mixed. The gap between complete li and li that an algorithm can list (Section 7) matches his distinction between complete li and our partial grasp of li.

Wang Fuzhi takes qi to be prior. This position corresponds to starting from the actual world, regarded as a structure W, and treating the li of the world as $\mathrm{Th}(W)$. Corollary 1 raises a difficulty for this position. If W is infinite, $\mathrm{Th}(W)$ has many non-isomorphic models, so the li read off from actual qi does not determine actual qi. A monist can reply that only the actual model is real and that the other models are mathematical artifacts with no claim to existence. The reply is coherent. However, the reply concedes that the identity of the actual qi cannot be recovered from the li of the actual qi, and this concession is significant for a view on which li is nothing over and above the order of qi.

T'oegye held that li issues, or acts, in the Four Beginnings. Satisfaction is a relation between a structure and a set of sentences, and model theory has no counterpart to a theory that acts on its models. T'oegye's position therefore cannot be expressed in this static framework, which marks a limit of the analogy (Section 11).

Yulgok held that only qi issues and that li rides on qi, and that li pervades while qi delimits. Both theses fit the framework without problems. The structure belongs to the model, the theory states what the model satisfies, and Section 9 shows that all variation occurs among the models. We do not take this fit as evidence that Yulgok was right, because the fit may only show that model theory treats principle as inert, as Yulgok did.

11 Objections and limits of the analogy

The most obvious objection to our framework is that the neo-Confucians did not think of li as a set of sentences. The structural reading of Section 3 answers this objection in part, because $\mathrm{Th}(M)$ is fixed by M and does not depend on any choice of axioms. But the response is only partial, because $\mathrm{Th}(M)$ still depends on the language $\mathcal{L}$ in which the sentences are written. One could remove this dependence by identifying li with the isomorphism type of M, since no language enters into an isomorphism type. However, that identification would make the "one li, many qi" problem trivial, because all realizations of an isomorphism type are isomorphic. But while languages have limitations on probing philosophy, they are crucial to discuss the potential differences between the li we can express and the actual li, as explored in Sections 5, 7 and 8.

There is another major objection concerning qi: the neo-Confucians regarded qi as constantly transforming, while classical model theory treats structures as static. While this objection cannot fundamentally be overcome, there are some ways to lessen the limitations, with model theory offering a way to describe growth of qi or of a model. An elementary end extension of a model of PA adds new elements above all existing elements while keeping the same sentences true, and the MacDowell-Specker theorem shows that every model of PA has a proper elementary end extension (MacDowell and Specker 1961). Any arithmetical qi can therefore grow without any change in the li that the qi satisfies, which recalls the neo-Confucian theme of endless generation (sheng sheng). End extensions only add new elements on top of a fixed structure, and a fuller account of change in qi would need other tools from logic, such as forcing extensions in set theory or Kripke models for temporal logic. Forcing builds a larger model from a given model of set theory, and the choice of construction can make a statement, such as the continuum hypothesis, that ZFC leaves undecided true or false in the larger model. Nevertheless, these remain limited, and more work is required to truly establish the formal parallel of dynamic qi.

The static character of model theory also explains why the framework has no counterpart

to T'oegye's active li (Section 10). Growth of the kind just described happens in the models while the theory stays fixed, so the framework can describe change in qi but not action by li.

12 Conclusion

In this paper, we provided a static hermeneutic framework for the relation between li and qi as the relation between a theory and its models. While we acknowledge its main limitation of being unable to probe dynamic qi, the framework separates the question whether li determines qi from the question whether li is prior to qi, with the framework mostly clarifying the former question. Model theory is largely silent on whether li is prior to qi as li (theory) and qi (model) exist simultaneously there.

The completeness theorem gives a formal counterpart to Zhu Xi's thesis that li and qi are neither separate nor mixed, because every consistent li is realized in some qi while a theory and a model remain different kinds of object.

Compactness guarantees that a li understood gradually and consistently in the investigation of things (gewu) cannot result in a wholly inconsistent understanding. In other words, we can reach the truth via finitary and gradual means.

A first-order li with an infinite qi always has multiple candidates for qi, and a li can fix an infinite qi only in a logic that gives up compactness or the Löwenheim-Skolem property. This lack of uniqueness offers an interpretation of why the incremental investigation of things might differ from the sudden penetration (huoran guantong) that Zhu Xi describes as its completion.

Morley's theorem and the standard examples show that the distribution of one li among many qi described by li yi fen shu can take quite different forms, depending on the li and on the size of qi.

Incompleteness separates the complete li of a structure from any li that an algorithm can list, which matches Zhu Xi's view that li is complete in each thing while human understanding

of li is partial.

Tennenbaum's theorem ties the uniqueness of qi to computability. If arithmetical qi is mechanical, the li of arithmetic picks out the standard natural numbers, and if qi is vital, one li again admits many qi. The limits set out in Section 8 make this result conditional and local, so the result does not pick whether mechanism or vitalism is true for qi.

The model-theoretic notion of a type gives a counterpart to the distinction between original and physical nature. All models of a complete theory share the same li, but each model realizes some types and omits others, much as turbid qi obscures li without removing li. Clear qi can correspond either to a model that realizes every type or to the prime model, which realizes only what li fixes, depending on what manifesting li is taken to mean. Yulgok's thesis that li pervades and qi delimits fits the same picture, since the theory is identical in every model and all particularity comes from the models.

The historical positions fit the framework to different degrees. Zhu Xi's logical priority of li corresponds to starting from a theory and then studying the model of the theory. Wang Fuzhi's priority of qi corresponds to reading the li from the actual world. If the world is infinite, that li does not determine the world up to isomorphism, so the identity of actual qi cannot be recovered from the li of actual qi. Yulgok's position has a formal counterpart, while T'oegye's active li has none, because satisfaction is a relation and a theory cannot act on its models. We note this as the main limitation left to future work.

Data availability, ethics and declaration of interests

This paper is purely theoretical and involves no data. The authors have no funding sources or conflicts of interest to declare. No experiment that requires ethical statements was performed.

References

Adler, Joseph A. 2023. "Re-forming Confucianism: Zhu Xi's Synthesis." In *The Oxford Handbook of Confucianism,* edited by Jennifer Oldstone-Moore. New York: Oxford University Press.

Angle, Stephen C., and Justin Tiwald. 2017. *Neo-Confucianism: A Philosophical Introduction.* Cambridge, UK: Polity Press.

Button, Tim, and Sean Walsh. 2018. *Philosophy and Model Theory.* Oxford: Oxford University Press.

Chan, Wing-tsit. 1963. *A Source Book in Chinese Philosophy.* Princeton, NJ: Princeton University Press.

Chang, C. C., and H. Jerome Keisler. 1990. *Model Theory.* 3rd ed. Studies in Logic and the Foundations of Mathematics 73. Amsterdam: North-Holland.

Chung, Edward Y. J. 1995. *The Korean Neo-Confucianism of Yi T'oegye and Yi Yulgok.* Albany, NY: State University of New York Press.

Dedekind, Richard. 1888. *Was sind und was sollen die Zahlen?* Braunschweig: Vieweg.

Gödel, Kurt. 1930. "Die Vollständigkeit der Axiome des logischen Funktionenkalküls." *Monatshefte für Mathematik und Physik* 37:349–360.

———. 1931. "Über formal unentscheidbare Sätze der Principia Mathematica und verwandter Systeme I." *Monatshefte für Mathematik und Physik* 38:173–198.

Godziszewski, Michał Tomasz, and Joel David Hamkins. 2017. "Computable Quotient Presentations of Models of Arithmetic and Set Theory." In *Logic, Language, Information, and Computation (WoLLIC 2017),* edited by Juliette Kennedy and Ruy J. G. B. de Queiroz, 140–152. Lecture Notes in Computer Science 10388. Berlin, Heidelberg: Springer. https://doi.org/10.1007/978-3-662-55386-2_10.

Henkin, Leon. 1950. "Completeness in the Theory of Types." *Journal of Symbolic Logic* 15 (2): 81–91.

Hodges, Wilfrid. 1993. *Model Theory.* Encyclopedia of Mathematics and Its Applications 42. Cambridge: Cambridge University Press.

Kalton, Michael C., Oaksook C. Kim, Sung Bae Park, Youngchan Ro, Wei-ming Tu, and Samuel Yamashita. 1994. *The Four-Seven Debate: An Annotated Translation of the Most Famous Controversy in Korean Neo-Confucian Thought.* Albany, NY: State University of New York Press.

Kim, Youngmin, and Youngyeon Kim. 2023. "Confucianism in Korea: Chosŏn Dynasty." In *The Oxford Handbook of Confucianism,* edited by Jennifer Oldstone-Moore. New York: Oxford University Press.

Lindström, Per. 1969. "On Extensions of Elementary Logic." *Theoria* 35 (1): 1–11.

Liu, JeeLoo. 2017. *Neo-Confucianism: Metaphysics, Mind, and Morality.* Hoboken, NJ: Wiley-Blackwell.

MacDowell, Robert, and Ernst Specker. 1961. "Modelle der Arithmetik." In *Infinitistic Methods: Proceedings of the Symposium on Foundations of Mathematics, Warsaw, 1959,* 257–263. Oxford and Warsaw: Pergamon Press / Państwowe Wydawnictwo Naukowe.

Marker, David. 2002. *Model Theory: An Introduction.* Graduate Texts in Mathematics 217. New York: Springer.

Morley, Michael. 1965. "Categoricity in Power." *Transactions of the American Mathematical Society* 114 (2): 514–538.

Pakhomov, Fedor. 2022. "How to Escape Tennenbaum's Theorem." *arXiv e-prints,* arXiv: 2209.00967 [math.LO].

Putnam, Hilary. 1980. "Models and Reality." *Journal of Symbolic Logic* 45 (3): 464–482.

Rabin, Michael O. 1958. "On Recursively Enumerable and Arithmetic Models of Set Theory." *Journal of Symbolic Logic* 23 (4): 408–416. https://doi.org/10.2307/2964015.

Ro, Young-chan. 1989. *The Korean Neo-Confucianism of Yi Yulgok.* Albany, NY: State University of New York Press.

Rosser, J. Barkley. 1936. "Extensions of Some Theorems of Gödel and Church." *Journal of Symbolic Logic* 1 (3): 87–91.

Shapiro, Stewart. 1991. *Foundations without Foundationalism: A Case for Second-Order Logic.* Oxford: Oxford University Press.

Shelah, Saharon. 1990. *Classification Theory and the Number of Non-Isomorphic Models.* 2nd ed. Studies in Logic and the Foundations of Mathematics 92. Amsterdam: North-Holland.

Shepherdson, John C. 1964. "A Non-Standard Model for a Free Variable Fragment of Number Theory." *Bulletin de l'Académie Polonaise des Sciences, Série des Sciences Mathématiques, Astronomiques et Physiques* 12:79–86.

Skolem, Thoralf. 1920. “Logisch-kombinatorische Untersuchungen über die Erfüllbarkeit oder Beweisbarkeit mathematischer Sätze.” *Skrifter utgitt av Videnskapsselskapet i Kristiania, I. Matematisk-naturvidenskabelig klasse,* no. 4, 1–36.

Tennenbaum, Stanley. 1959. “Non-Archimedean Models for Arithmetic.” *Notices of the American Mathematical Society* 6:270.

Wang Fuzhi. 2011. *Du Sishu daquan shuo* [Commentary on the Great Compendium of the Four Books]. Chuanshan quanshu. Changsha: Yuelu shushe.

Zhu Xi. (1270) 1986. *Zhuzi yulei* [Classified Conversations of Master Zhu]. Edited by Li Jingde. Beijing: Zhonghua shuju.